

Proofs and Problem Solving

Week 8: Relations and Functions

Notes based on Martin Liebeck's

A Concise Introduction to Pure Mathematics

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Week 8: Relations and Functions

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13. More set theory and equivalence relations

Week 1 introduced us to the notion of sets – fundamental mathematical objects that allow us to group together objects that share some pre-defined property. *Relations* give us a way to express how, or when, two objects share a particular relationship. We’ve already dealt with lots of specific relations in PPS so far. For example, the relation “ $<$ ” on $\mathbb{R}$ allows us to express when one real number x is smaller than another, y , by writing $x < y$. As with “ $<$ ”, it is possible for some relations to only hold in one direction (e.g. $2 < 3$ but $3 \not< 2$) and this direction is indicated by our expression of a pair, x and y , of related elements as an *ordered pair* (x, y) .

Functions (or maps) are special relations on a pair of sets (the *domain* and the *codomain*) with the added restriction that each element of the domain must only be the first element in a related pair exactly once. We’ve also dealt with lots of specific functions in PPS so far. For example, the modulus function “ $|\cdot|$ ” on the set $\mathbb{R}$ relates every $x \in \mathbb{R}$ to itself if $x \geq 0$ and to $-x$ if $x < 0$. The modulus function “ $|\cdot|$ ” from $\mathbb{C}$ to $\mathbb{R}$ relates every complex number to its distance from the origin when drawn on the Argand diagram.

13.1 Cartesian products

In coordinate geometry, you’re used to the idea of an *ordered pair* (x, y) . The defining feature of ordered pairs is that $(x, y) = (u, v)$ if and only if $x = u$ and $y = v$. The word “ordered” signals that (x, y) and (y, x) are different, unless $x = y$. For instance, $(1, 2)$ and $(2, 1)$ are different points of the plane. In contrast, the *sets* $\{x, y\}$ and $\{y, x\}$ are equal.

In these examples, x and y are both elements of the set $\mathbb{R}$. But we can talk about ordered pairs for any sets; and x and y don’t even need to belong to the *same* set. So, given any element x of a set X , and any element y of a set Y , we can form the ordered pair (x, y) . Again, the defining feature of ordered pairs is that

$$(x, y) = (u, v) \iff x = u \text{ and } y = v$$

for $x, u \in X$ and $y, v \in Y$.

Definition 13.1 Let X and Y be sets. Their *cartesian product* is the set of ordered pairs

$$X \times Y = \{(x, y) : x \in X, y \in Y\}.$$

We can take products of very different sets. For example, $\{1, 2, \dots, 8\} \times \{\text{Cat}, \text{Dog}\}$ is the set of all pairs of numbers between 1 and 8 with either a cat or dog, so it has elements like $(2, \text{Dog})$, $(8, \text{Cat})$, etc.

We can also take products of the same set. For instance,

$$\mathbb{R} \times \mathbb{R} = \mathbb{R}^2.$$

A rectangle in $\mathbb{R}^2$ can be defined using two intervals, $[a, b]$ in the x -axis and $[c, d]$ in the y -axis:

$$[a, b] \times [c, d] = \{(x, y) : x \in [a, b], y \in [c, d]\}.$$

If you don't see why, draw a picture!

Given sets $X_1, X_2, \dots, X_n$, we can also consider *ordered n -tuples* $(x_1, x_2, \dots, x_n)$, where $x_1 \in X_1, x_2 \in X_2, \dots, x_n \in X_n$. The word 'ordered' tells us that

$$(x_1, x_2, \dots, x_n) = (u_1, u_2, \dots, u_n) \iff x_1 = u_1, x_2 = u_2, \dots, x_n = u_n$$

$(x_i, u_i \in X_i)$. The set of all ordered n -tuples $(x_1, \dots, x_n)$ is written as $X_1 \times X_2 \times \dots \times X_n$.

13.2 Maps

A *map* (or *mapping* or *function*) consists of three things:

- a set X , called the *domain* of the map;
- a set Y , called the *codomain* of the map;
- a rule that assigns to each element $x \in X$ an element $f(x) \in Y$.

We write such a map as $f: X \rightarrow Y$.

In order for two maps to be equal, *they must have the same domain and codomain* as well as the same rule.

Example 13.1 There are functions

$$f: \mathbb{Z} \rightarrow \mathbb{Z}, \quad g: \mathbb{Z} \rightarrow \mathbb{R}, \quad h: \mathbb{R} \rightarrow \mathbb{R}$$

defined by

$$f(x) = x^2 \ (x \in \mathbb{Z}), \quad g(x) = x^2 \ (x \in \mathbb{Z}), \quad h(x) = x^2 \ (x \in \mathbb{R}).$$

Although f , g and h are all given by the same formula, *they are all different functions*. For instance, $f \neq g$ because f and g have different codomains. The domain and codomain are part of the function!

Another important point is that each input value $x \in X$ of a function $f: X \rightarrow Y$ gives *exactly one* output value $f(x)$.

Example 13.2 None of the following are a valid definition of a function:

- $F: [0, \infty) \rightarrow \mathbb{R}$ defined by $F(x) = \pm\sqrt{x}$;
- $G: \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $G(x) = x/2$;
- $H(x) = 2x$.

The problem with F is that it assigns *two* values to some elements of the domain (in fact, every element of the domain apart from 0).

The problem with G is that, since half of an odd integer is not an integer, the formula for $G(x)$ does not define any element of the codomain when x is odd.
 The problem with H is that the domain and codomain have not been specified.

If we want to be even more precise, we can define a *map* from X to Y as a subset $G \subseteq X \times Y$ such that for each $x \in X$, there is exactly one element $y \in Y$ satisfying $(x, y) \in G$. Then

$$G = \{(x, y) \in X \times Y : y = f(x)\}$$

of $X \times Y$. We call G the *graph* of f . But in practice, no one ever says things like " $(\frac{\pi}{2}, 1) \in \sin$ ", which is what this definition suggests. We say $\sin(\frac{\pi}{2}) = 1$, as usual.

Example 13.3 Let n be a natural number. Recall that $\mathbb{Z}/n$ is the set $\{0, 1, \dots, n-1\}$. Then we may define a map $\mu_n: \mathbb{Z} \rightarrow \mathbb{Z}/n$ by the rule that for any integer m ,

$$\mu_n(m) = m \bmod n.$$

In other words, $\mu_n(m)$ is the remainder of m after dividing by n .

Sometimes, it's handy to use the following notation: we may define a map $f: X \rightarrow Y$ by the assignment

$$x \mapsto [\text{some formula involving } x].$$

Example 13.4 We may define a map $e: \mathbb{R} \rightarrow \mathbb{R}$ by the rule

$$x \mapsto \exp(-x^2).$$

This is another way of saying that $e(x) = \exp(-x^2)$.

Don't get the arrows $\rightarrow$ and $\mapsto$ mixed up! The $\rightarrow$ arrow goes between sets, as in $f: X \rightarrow Y$ and read as "to". The $\mapsto$ arrow goes between elements, as in $\sin: \frac{\pi}{2} \mapsto 1$ and is read as "is mapped to". For example, " $n \rightarrow n^2$ " is wrong.

Example 13.5 Let Y be a set and let $X \subseteq Y$ be a subset. There is a map $i: X \rightarrow Y$ given by the rule $i(x) = x$. This is called the *inclusion map*. When $X = Y$, the inclusion map $i: X \rightarrow X$ is called the *identity map* $\text{id}: X \rightarrow X$.

Given maps $f: X \rightarrow Y$ and $g: Y \rightarrow Z$, you can *do f first, and then g* . That is, one can form a new map $g \circ f: X \rightarrow Z$ such that for each $x \in X$,

$$(g \circ f)(x) = g(f(x)) \in Z.$$

The map $g \circ f$ is called the *composite* of f and g . It is easy enough to see that $g \circ \text{id} = g$ and $\text{id} \circ f = f$. Moreover, the process of composition is *associative*, meaning that $(h \circ g) \circ f = h \circ (g \circ f)$.

13.3 Injectivity and surjectivity

Definition 13.2 Let $f: X \rightarrow Y$ be a function. We say that

- the map f is *injective* (or an *injection* or *one-to-one*) if and only if

$$(\forall x, x' \in X)(f(x) = f(x') \implies x = x');$$

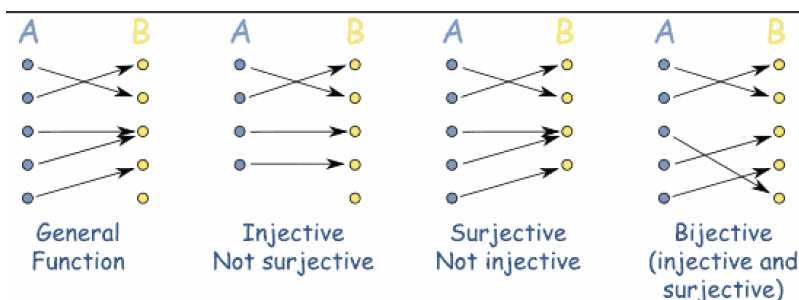
- the map f is *surjective* (or a *surjection* or *onto*) if and only if

$$(\forall y \in Y)(\exists x \in X)(f(x) = y);$$

- the map f is *bijective* (or a *bijection* or a *one-to-one correspondence*) if and only if it is injective and surjective.

Here's another way to say it:

- f is injective if for each $y \in Y$, there is *at most* one element $x \in X$ such that $f(x) = y$;
- f is surjective if for each $y \in Y$, there is *at least* one element $x \in X$ such that $f(x) = y$;
- f is bijective if for each $y \in Y$, there is *exactly* one element $x \in X$ such that $f(x) = y$.



When proving that a function $f: X \rightarrow Y$ is injective, start by assuming $x, x' \in X$ are such that $f(x) = f(x')$ and show that this implies $x = x'$. To show a function is not injective, you just need to find one pair $x \neq x'$ in X with $f(x) = f(x')$.

Example 13.6 Claim: The function $f: \mathbb{N}_0 \rightarrow \mathbb{N}_0$ defined by $f(n) = n^2$ is injective.

Proof: Let $m, n \in \mathbb{N}_0$ be such that $f(m) = f(n)$. Then $m^2 = n^2$, and so

$$0 = m^2 - n^2 = (m - n)(m + n).$$

If $m + n = 0$ then $m = 0$ and $n = 0$, so $m = n$. Otherwise, $m - n = 0$, so $m = n$. Thus f is injective. $\square$

It is not surjective, since there is no n such that $f(n) = 2$. If we change the domain to $\mathbb{Z}$, then $f: \mathbb{Z} \rightarrow \mathbb{N}_0$ is no longer injective, as $f(-1) = f(1)$.



When proving that a function $f: X \rightarrow Y$ is surjective, start with “Let $y \in Y$ ” and show that there must be an $x \in X$ such that $f(x) = y$. To show it is not surjective, you just need to find one $y \in Y$ such that there is no $x \in X$ satisfying $f(x) = y$.

Example 13.7 Claim: The function $g: \mathbb{R} \rightarrow [-5, +\infty)$ defined by $g(x) = x^2 + 2x - 4$ is surjective.

Proof: Let $y \geq -5$. We need to show there is some $x \in \mathbb{R}$ such that $g(x) = y$, so $x^2 + 2x - 4 = y$. By the quadratic formula, that means

$$x = \frac{-2 \pm \sqrt{4 + 4(4 + y)}}{2}$$

and since $y \geq -5$, the square root is defined and this gives a solution. Thus, since we

can find such an $x \in \mathbb{R}$ for every $y \in [-5, \infty)$, this implies that g is surjective. $\square$

If we change the codomain of g to $[-6, +\infty) = \{x \in \mathbb{R} : x \geq -6\}$, then the new g is not surjective: if $-6 = g(x) = x^2 + 2x - 4$ for some $x \in \mathbb{R}$, then the quadratic formula says that $x = -1 \pm i$, but these roots are not in the real line. Thus, there is no x such that $g(x) = -6$, and so $g: \mathbb{R} \rightarrow [-6, +\infty)$ is not surjective.

The proof of the following result is a good one to study, since it showcases how to prove injectivity and surjectivity.

Theorem 13.1 Let $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ be functions.

1. If f and g are injective, so is $g \circ f$.
2. If f and g are surjective, so is $g \circ f$.
3. If f and g are bijective, so is $g \circ f$.

Proof. 1. Suppose f and g are injective. To show that $g \circ f$ is injective, we need to show that if $(g \circ f)(x) = (g \circ f)(x')$ (with $x, x' \in X$), then $x = x'$. So, suppose $x, x' \in X$ satisfy $(g \circ f)(x) = (g \circ f)(x')$. Then $g(f(x)) = g(f(x'))$. Since g is injective, this means $f(x) = f(x')$. Since f is injective, this implies $x = x'$, as desired. This completes the proof that $g \circ f$ is injective.

2. Now suppose f and g are surjective. To show that $g \circ f$ is surjective, we need to show that for all $z \in Z$, there is some $x \in X$ such that $z = (g \circ f)(x)$. Let $z \in Z$. Since g is surjective, there is some $y \in Y$ such that $z = g(y)$. Since f is surjective, there is some $x \in X$ such that $f(x) = y$. Then

$$(g \circ f)(x) = g(f(x)) = g(y) = z,$$

as desired. This finishes the proof that $g \circ f$ is surjective.

3. If g and f are bijective, then they are both injective and surjective, so by the previous two parts, $g \circ f$ is also both injective and surjective, hence $g \circ f$ is bijective. $\blacksquare$

When mathematicians are presented with a set, they often won't be interested in what the elements happen to be named. (Question: are the sets $\{1, 2, 3, \dots\}$ and $\{I, II, III, \dots\}$ the same or different?)

This is why bijections are important. A bijection between sets X and Y is a *relabelling* of the elements (for instance, relabelling 1 as I, and 2 as II, and so on). That relabelling is a perfect match: you never use the same label twice, and all the labels are used. That is, a bijection $f: X \rightarrow Y$ is a map such that for each $y \in Y$, there is a unique $x \in X$ satisfying $f(x) = y$.

Example 13.8 Let S and T be two sets. There is a bijection $\sigma: S \times T \rightarrow T \times S$ given by $\sigma(s, t) = (t, s)$.

Let $f: X \rightarrow Y$ be a bijection. For each $y \in Y$, there is one and only one element $x \in X$ such that $f(x) = y$. We can give this element x a name: call it $g(y)$. But now we have produced a rule that assigns to each element $y \in Y$ an element $g(y) \in X$ so we have defined a function $g: Y \rightarrow X$. It has the properties that

$$g \circ f = \text{id} \quad \text{and} \quad f \circ g = \text{id}.$$

We call the map $g: Y \rightarrow X$ the *inverse* of $f: X \rightarrow Y$, and write g as f^{-1} .

What we've just shown is that every bijection has an inverse. Conversely, you can show that any map f with an inverse (meaning, a function $g: Y \rightarrow X$ such that $g \circ f = \text{id}$ and $f \circ g = \text{id}$) must be bijective. So: 'bijective' is equivalent to 'invertible'.

13.4 Images and preimages

Definition 13.3 Let $f: X \rightarrow Y$ be a map of sets. For $A \subseteq X$, the *image* of A under f is the subset

$$f(A) = \{f(x) : x \in A\} \subseteq Y.$$

For $B \subseteq Y$, the *preimage* of B under f is the subset

$$f^{-1}(B) = \{x \in X : f(x) \in B\} \subseteq X.$$

Example 13.9 If $f: \mathbb{N} \rightarrow \mathbb{N}$ is given by $f(n) = n + 1$, and $A = \{2n : n \in \mathbb{N}\}$ is the set of even natural numbers, then the image of A under f is

$$f(A) = \{f(n) : n \in A\} = \{f(2n) : n \in \mathbb{N}\} = \{2n + 1 : n \in \mathbb{N}\}.$$

That is, $f(A)$ is the set of the odd integers ≥ 3 .

Some authors call the preimage the *inverse image*.

Warning: The preimage $f^{-1}(B)$ is defined for *every* map f , whether or not f has an inverse! We're not giving a meaning to " f^{-1} " in isolation. We're only giving a meaning to " $f^{-1}(B)$ ".

If f *does* have an inverse (that is, if f is bijective) then the notation $f^{-1}(B)$ is in principle ambiguous. One person might interpret it as the preimage of B under the function f , and another might interpret it as the image of B under the function f^{-1} . But in fact, these two sets are the same (why?), so there is no ambiguity.

If you ever see an expression like " $f^{-1}(y)$ ", where $f: X \rightarrow Y$ and y is an *element* of Y , then it must be referring to f^{-1} as a function, so the expression only makes sense if f has an inverse. But an expression like " $f^{-1}(B)$ ", where B is a *subset* of Y , always makes sense, regardless of whether f has an inverse.

Theorem 13.2 Let $f: X \rightarrow Y$ be a function and $B, B' \subseteq Y$. Then

$$f^{-1}(B \cap B') = f^{-1}(B) \cap f^{-1}(B')$$

and

$$f^{-1}(B \cup B') = f^{-1}(B) \cup f^{-1}(B')$$

Proof. To show the first equality, note that $x \in f^{-1}(B \cap B')$ if and only if $f(x) \in B \cap B'$. Therefore, $f(x) \in B$ and $f(x) \in B'$, which is true if and only if $x \in f^{-1}(B)$ and $x \in f^{-1}(B')$, that is, if and only if $x \in f^{-1}(B) \cap f^{-1}(B')$. Thus, $f^{-1}(B \cap B') = f^{-1}(B) \cap f^{-1}(B')$.

The second equation has a similar proof (basically just change "and" to "or"). ■

13.5 Comparing sizes of sets with functions

In the diagrams earlier, note that whenever the function $f: X \rightarrow Y$ is injective there have to be at least as many elements in Y as there are in X , otherwise the arrows from X couldn't point to distinct points in Y . Similarly, when the function is surjective, there need to be at least as many points in X as there are in Y , since for each point in Y there is at least one point in X that maps to it. We make these observations precise in the following theorem. For a finite set A , we will let $|A|$ equal the size of A , that is, the number of elements in A .

Theorem 13.3 Let X and Y be finite sets and let $f : X \rightarrow Y$ be a function.

1. If f is injective, then $|X| \leq |Y|$.
2. If f is surjective, then $|X| \geq |Y|$.
3. If f is bijective, then $|X| = |Y|$.

Proof. 1. Since X is finite, we can list its elements as $x_1, \dots, x_n$ where $n = |X|$. If f is injective, then the elements $f(x_1), \dots, f(x_n)$ are distinct, so there are exactly n elements in the set $B = \{f(x_1), \dots, f(x_n)\}$. Since this is a subset of Y , the set Y must have at least as many elements as B , so $|Y| \geq n = |X|$.

2. Since Y is finite, we can list its elements as $y_1, \dots, y_m$ where $m = |Y|$. If f is surjective, then for each $i = 1, \dots, m$, we can choose some $x_i \in X$ such that $f(x_i) = y_i$. The elements $x_1, \dots, x_m$ of X are distinct, since if $x_i = x_j$ then $f(x_i) = f(x_j)$, i.e. $y_i = y_j$, hence $i = j$. Thus, X has at least m distinct elements, giving $|X| \geq m = |Y|$.

3. This follows by combining (1) and (2). ■

13.6 Equivalence relations

Loosely speaking, a relation is a way for two things to be connected to each other. The relation “=” is an example: two things are related via “=” if they are equal. Another example is the relationship that one number is at most another, and we write $a \leq b$ when we want to say a and b are related in this way. A less mathematical example is the relation “daughter of”, so for example “ a is the daughter of b ”.

Definition 13.4 A *relation* R on a set X is a subset of $X \times X$. Given $a, b \in X$, we write aRb if $(a, b) \in R$.

This may seem like a strange way of defining relations, but this definition encapsulates what we do when we define a relation: a relation is a specific way of pairing up elements of a set. In the earlier example, the daughter-of relation is the set R of all pairs of people (a, b) where a is the daughter of b . Another familiar example is $\leq$: we declare $a \leq b$ if the pair (a, b) is among the set R of pairs where a is at most b .

Here are some important properties relations can have:

Definition 13.5 Let X be a set. Let $\sim$ be a relation on X . We say $\sim$ is

- *reflexive* if for all $a \in X$, we have $a \sim a$;
- *symmetric* if for all $a, b \in X$, we have $a \sim b \implies b \sim a$;
- *transitive* if for all $a, b, c \in X$, we have $a \sim b \sim c \implies a \sim c$.

If $\sim$ satisfies all three of these properties, we say that $\sim$ is an *equivalence relation*.

The relations we discussed earlier satisfy different combinations of these properties.

Example 13.10 Equality is an equivalence relation. We also showed in Week 7 that $\equiv \pmod{m}$ is an equivalence relation.

$\leq$ is reflexive and transitive but not symmetric ($1 \leq 2$ but $2 \not\leq 1$).

The daughter-of relation is not symmetric: if a is the daughter of b , then b is certainly not the daughter of a , and it is certainly not reflexive since no one is the daughter of themselves. The daughter-of relation is also not transitive (if a is the daughter of b and b is the daughter of c , that does not mean a is the daughter of c).

Relations do not necessarily have to satisfy all or any of the three properties above. So whenever we define a relation, we need to check these properties carefully.



Disproving properties: To disprove a property like symmetry, it suffices to just find *one* pair a and b for which $a \sim b$ but $b \not\sim a$. In the previous example, clearly $\leq$ fails the symmetry property for more than just the numbers 1 and 2, but to verify that symmetry fails, we only need to show it fails for one pair.

Why care? Relations and equivalence relations come about when there is a collection of objects and you would like to treat many of them as really the same object. For example, a rational number is a fraction $\frac{p}{q}$ where $p \in \mathbb{Z}$ and $q \in \mathbb{N}$. Notice that the notation $\frac{p}{q}$ is really just an ordered pair, which we could have also write as (p, q) . But we know that several fractions should give the same number, e.g. $\frac{2}{1} = \frac{4}{2}$. Really what we're doing here is defining an equivalence relation $\sim$ on $\mathbb{Z} \times \mathbb{N}$ by

$$(p, q) \sim (r, s) \iff ps = qr.$$

(The point is that $ps = qr \iff p/q = r/s$.) You'll see other important examples in future courses.

Definition 13.6 Let $\sim$ be an equivalence relation on a set X , and let $x \in X$. The *equivalence class* of x is

$$\text{cl}(x) = \{y \in X : x \sim y\}.$$

The set of *equivalence classes* is the set $X/\sim = \{\text{cl}(x) : x \in X\}$. This set comes with a map $\text{cl}: X \rightarrow X/\sim$.

The map cl is a surjection: every equivalence class is the equivalence class of *something*.

Example 13.11 Consider the relation on people that $a \sim b$ if a and b were born in the same country. One can check this is an equivalence relation. What if we want to describe the equivalence classes? We could just say "they are the sets $\text{cl}(x)$ where x ranges over all people" but this is not helpful in general; this says no more about the relation or what the classes look like than making the same statement about any other relation. A more natural way of describing the equivalence classes is that they are just the sets of people born in a given country, that is, they are the sets A_x of people born in country x as x ranges over all countries. So, equivalence classes correspond one-to-one with countries.

Note that in the previous example, the relation partitioned people into disjoint groups (in this case, based on their country of birth). This happens with all equivalence relations.

Example 13.12 Let $m \in \mathbb{N}$, and consider the equivalence relation on $\mathbb{Z}$ defined by $\equiv \pmod{m}$. What are the equivalence classes? If $x \in \mathbb{Z}$, then the equivalence class of x is

$$\begin{aligned} \text{cl}(x) &= \{y \in \mathbb{Z} : x \equiv y \pmod{m}\} \\ &= \{y \in \mathbb{Z} : y = x + mj \text{ for some integer } j\} \\ &= \{x + mj : j \in \mathbb{Z}\}. \end{aligned}$$

Notice that for each x , there is some $i \in \{0, 1, \dots, m-1\}$ such that $x \equiv i \pmod{m}$, and then $\text{cl}(x) = \text{cl}(i)$. Thus, all equivalence classes are one of

$$\text{cl}(0), \text{cl}(1), \dots, \text{cl}(m-1).$$

So the equivalence classes are the arithmetic progressions of the form $\dots, i - 2m, i - m, i, i + m, i + 2m, \dots$ for some $i \in \{0, 1, \dots, m-1\}$. Observe that $i \mapsto \text{cl}(i)$ defines a bijection from $\mathbb{Z}/m = \{0, 1, \dots, m\}$ to the set $\{\text{cl}(0), \dots, \text{cl}(m-1)\}$. In this way, we think of $\mathbb{Z}/m$ as the “set of integers, modulo m .”

13.7 Exercises

The relevant exercises in Liebeck’s book are in Chapter 19.

■ **Exercise 13.1** Consider the following functions. Say which composites are defined (e.g. is $f_2 \circ f_1$ defined? Is $f_4 \circ f_6$?) Then decide whether each of the composites is injective, surjective, both, or neither. Do any of them have an inverse function?

- (a) $f_1: \mathbb{R} \rightarrow \mathbb{Z}$ defined by $f_1(x) = \lfloor x \rfloor$
- (b) $f_2: \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $f_2(x) = x + 1$
- (c) $f_3: \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $f_3(x, y) = xy$
- (d) $f_4: \mathbb{R} \rightarrow \mathbb{Z} \times \mathbb{Z}$ defined by $f_4(x) = (\lfloor x \rfloor, \lceil x \rceil)$
- (e) $f_5: \{1, \dots, 10\} \rightarrow \{0, 1\}$ defined by $f_5(x) = \begin{cases} 0 & \text{if } x \text{ is even} \\ 1 & \text{if } x \text{ is odd} \end{cases}$
- (f) $f_6: \{0, 1\} \rightarrow \{1, \dots, 10\}$ defined by $f_6(0) = 10, f_6(1) = 7$

■ **Exercise 13.2** Let $f: X \rightarrow Y$ be a function.

- (a) Show that f is injective if and only if for all sets Z and functions $g, h: Z \rightarrow X$, $f \circ g = f \circ h$ implies $g = h$.
- (b) Show that f is surjective if and only if for all sets Z and functions $g, h: Y \rightarrow Z$, $g \circ f = h \circ f$ implies $g = h$.

■ **Exercise 13.3** Let $f: X \rightarrow Y$.

- (a) Show that $f^{-1}(f(A)) \supseteq A$ for all $A \subseteq X$, and show that these two sets can be unequal.
- (b) Show that $f^{-1}(f(A)) = A$ for all $A \subseteq X$ if and only if f is injective.

■ **Exercise 13.4** Let $f: X \rightarrow Y$.

- (a) Show that $f(f^{-1}(B)) \subseteq B$ for all $B \subseteq Y$, and give an example of an $f: X \rightarrow Y$ and a set $B \subseteq Y$ where $f(f^{-1}(B)) \neq B$.
- (b) Show that $f(f^{-1}(B)) = B$ for all $B \subseteq Y$ if and only if f is surjective.

■ **Exercise 13.5** Let X and Y be finite sets, $m \in \mathbb{N}$, and suppose $f: X \rightarrow Y$ is a function such that for all $y \in Y$, $|f^{-1}(\{y\})| = m$. Prove that

$$|X| = m \cdot |Y|.$$

■ **Exercise 13.6** Let $f: X \rightarrow Y$ be a function and $A, B \subseteq X$. Show that $f(A \cup B) = f(A) \cup f(B)$. Give an example of when we can have $f(A \cap B) \neq f(A) \cap f(B)$.

■ **Exercise 13.7** Let $f: X \rightarrow Y$ be a function. For $a, b \in X$, declare $a \sim b$ if $f(a) = f(b)$. Show that $\sim$ defines an equivalence relation on X .

■ **Exercise 13.8** Let $f: X \rightarrow Y$ and $\sim$ be as in the previous problem. For $y \in f(X)$, define $F(y) = f^{-1}(\{y\})$ (where recall $f^{-1}(A) = \{x \in X: f(x) \in A\}$). Show that F is a bijection between

$f(X)$ and the set of equivalence classes in X under the relation $\sim$. *Hint: As a first step, you'll need to show the values of F are indeed equivalence classes.*

■ **Exercise 13.9** Show that $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^{2n+1}$ is bijective for $n \in \mathbb{N}$.

■ **Exercise 13.10 Quite Challenging!** Find bijections between the following sets:

- (a) $\mathbb{R}$ and $(0, \infty)$
- (b) $(0, \infty)$ and $(1, \infty)$
- (c) (**Even more challenging!**) $(0, \infty)$ and $[0, \infty)$
- (d) $(0, \infty)$ and $(0, 1)$.

■ **Exercise 13.11** The following argument supposedly shows that the reflexivity condition is unnecessary for a nonempty relation $\sim$, that is, it can be derived from symmetry and transitivity:

“Suppose $a \sim b$. By symmetry, $b \sim a$. Since $a \sim b$ and $b \sim a$, by transitivity, $a \sim a$. Therefore, $\sim$ is reflexive.”

What's wrong with this argument?

■ **Exercise 13.12** Determine which of the following relations $\sim$ on the given set S are reflexive, symmetric, or transitive.

1. S is the set of lines in $\mathbb{R}^2$ and $a \sim b$ if they intersect.
2. $S = \mathbb{C} \setminus \{0\}$ and $x \sim y$ if x and y are contained in the same line passing through the origin.
3. S is the set of subsets of $\mathbb{R}$ and $A \sim B$ if $A \subseteq B$.

14. Applications: What's bigger than ∞ ?*

This section is not required reading, but it is a preview into another aspect of mathematics that you will learn about later: cardinality

Consider the following question: *How do we compare sizes of infinite sets?* The sets $\mathbb{N}$, $\mathbb{Q}$, and $\mathbb{R}$ are all infinite, but is any one *more infinite* than the other?

We first need a notion of size. Recall that in Theorem 13.3 we showed how two finite sets that have a bijection between them have the same size. We use this idea to define when two infinite sets have the same size:

Definition 14.1 We say two sets A and B *have the same cardinality*, or are *isomorphic*, if there is a bijection $f: A \rightarrow B$. We write $|A| = |B|$ or $A \cong B$. If there is an injective map $f: A \rightarrow B$, we write $|A| \leq |B|$, and if $|A| \leq |B|$ and there is no injective map from B to A , we write $|A| < |B|$.

When $|A| \leq |\mathbb{N}|$, we say A is *countable*.

(This is the usual modern definition, but Liebeck's book defines A to be countable if $|A| = |\mathbb{N}|$. So for him, finite sets are not countable, and for us, they are.)

■ **Exercise 14.1** Show that the relation $\sim$ is an equivalence relation on the collection of sets.

If A is countable and infinite then $A \cong \mathbb{N}$, in which case there is a bijection $f: \mathbb{N} \rightarrow A$ and

$$A = \{f(1), f(2), \dots\}.$$

So in a countable set (whether finite or infinite), you can count off the elements of A as a list.

Example 14.1 The set $\mathbb{Z}$ is countable. Define a map $f: \mathbb{Z} \rightarrow \mathbb{N}$ by

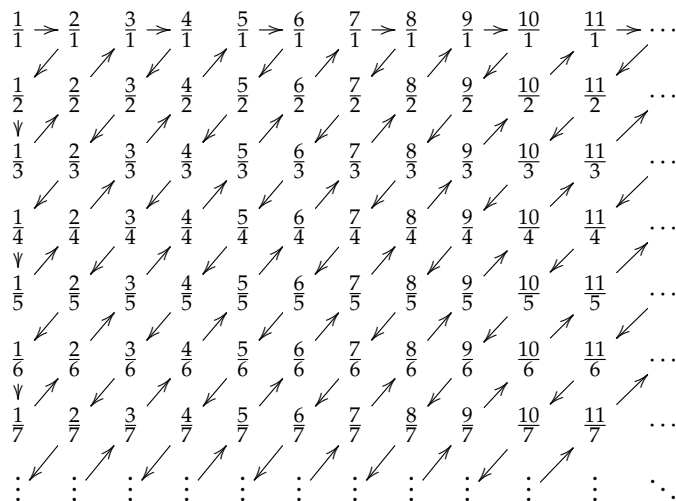
$$f(n) = \begin{cases} 2n & \text{if } n \geq 0, \\ -2n - 1 & \text{if } n < 0. \end{cases}$$

You can check that f is a bijection, and so $\mathbb{Z} \cong \mathbb{N}$...even though you might think of $\mathbb{Z}$ as “bigger” than $\mathbb{N}$!

More surprising is the following:

Theorem 14.1 $\mathbb{Q} \cong \mathbb{N}$.

Proof. The proof of this uses the “zig-zag” trick: write all the rational numbers in an infinite array, and then start listing all the rationals by going up and down each diagonal as shown:



So the list would be:

$$\frac{1}{1}, \frac{2}{1}, \frac{1}{2}, \frac{3}{1}, \frac{2}{2}, \frac{4}{1}, \frac{3}{2}, \frac{2}{3}, \frac{5}{1}, \frac{4}{2}, \frac{3}{3}, \frac{6}{1}, \frac{5}{2}, \frac{4}{3}, \frac{3}{4}, \frac{7}{1}, \frac{6}{2}, \frac{5}{3}, \frac{4}{4}, \frac{3}{5}, \frac{2}{6}, \frac{8}{1}, \frac{7}{2}, \frac{6}{3}, \frac{5}{4}, \frac{4}{5}, \frac{3}{6}, \frac{2}{7}, \frac{9}{1}, \frac{8}{2}, \frac{7}{3}, \frac{6}{4}, \frac{5}{5}, \frac{4}{6}, \frac{3}{7}, \frac{2}{8}, \frac{1}{9}, \frac{10}{1}, \frac{9}{2}, \frac{8}{3}, \frac{7}{4}, \frac{6}{5}, \frac{5}{6}, \frac{4}{7}, \frac{3}{8}, \frac{2}{9}, \frac{1}{10}, \dots$$

Note that this list repeats numbers (e.g. the number 1 appears as both $\frac{1}{1}$ and $\frac{2}{2}$). But we can just remove numbers that repeat, so the modified list looks like this:

$$\frac{1}{1}, \frac{2}{1}, \frac{3}{1}, \frac{4}{1}, \frac{5}{1}, \frac{6}{1}, \frac{7}{1}, \frac{8}{1}, \frac{9}{1}, \frac{10}{1}, \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \frac{7}{2}, \frac{9}{2}, \frac{1}{3}, \frac{2}{3}, \frac{4}{3}, \frac{6}{3}, \frac{8}{3}, \frac{10}{3}, \frac{1}{4}, \frac{3}{4}, \frac{5}{4}, \frac{7}{4}, \frac{9}{4}, \frac{1}{5}, \frac{2}{5}, \frac{4}{5}, \frac{6}{5}, \frac{8}{5}, \frac{10}{5}, \dots$$

This gives a bijection from $\mathbb{N}$ to $\mathbb{Q}$, where we send $n \in \mathbb{N}$ to the n th term in this list. ■

Not everything is countable though:

Theorem 14.2 Let S be a set and let $P(S)$ denote its *power set*, consisting of the subsets of S , i.e. $P(S) = \{A : A \subseteq S\}$. Then $|S| < |P(S)|$.

■ **Exercise 14.2** Prove the above theorem. *Hint: Suppose there is a bijection $f : S \rightarrow P(S)$ and think about the set $B = \{x \in S : x \notin f(x)\}$.*

It can also be shown that $\mathbb{R} \cong P(\mathbb{N})$, so $|\mathbb{N}| < |\mathbb{R}|$. That is, $\mathbb{R}$ is uncountable.

A natural question arises. We have seen that $|\mathbb{N}| < |\mathbb{R}|$, but is there anything in between? This is the *continuum hypothesis* posed by George Cantor:

Continuum Hypothesis (CH) There is no set S with $|\mathbb{N}| < |S| < |\mathbb{R}|$.

Set theory often uses a collection of axioms called *Zermelo–Fraenkel with choice* (ZFC; other axiom systems are available). In 1940, Kurt Gödel showed that the CH cannot be

disproved assuming ZFC, so it would seem then that CH has to be true... but then Paul Cohen showed in 1963 that CH cannot be *proved* assuming ZFC. In other words, CH is *independent* of ZFC: one can assume it's true or false and it won't contradict anything proved using the ZFC axioms!

